What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the [Università dell'Insubria]() in Como (Italy). Please direct questions and suggestions to [stefano.covino@inaf.it]().

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TIME-DOMAIN ASTROPHYSICS

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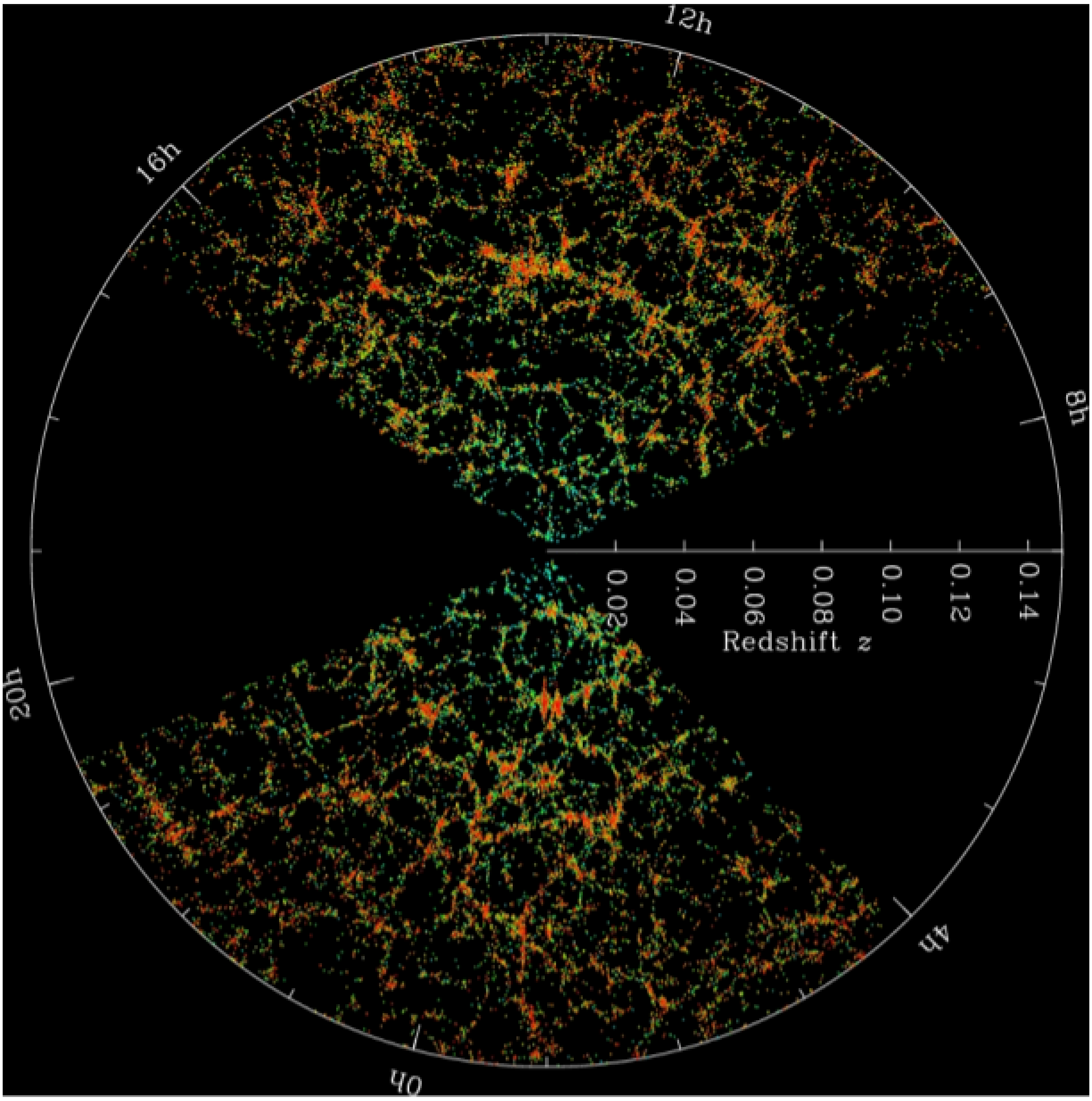
21st Century Challenges

- The expression "big data" in modern astronomy is not just a “hot keyword”.
- Observational astronomy today is a considerable enterprise with billions of dollars supporting ~20,000 scientists producing ~15,000 refereed papers annually.
- Not to mention the theoretical efforts, often based on profitable synergies with other fields.
- Astrostatistics is playing an increasing role in the analysis of astronomical observations and linking data to astrophysical theory.

Open problems in astrostatistics

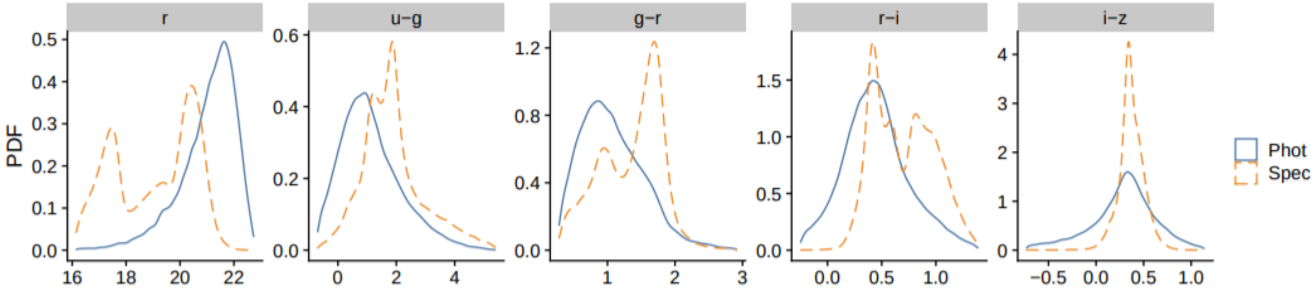
Galaxy clustering and large-scale structure

- Galaxy clustering: the distribution of galaxies in space proves to be surprising complex from the viewpoint of spatial point processes.
- Many statistical studies of large-scale structure rely on isotropic two- and three-point correlation functions as well as Fourier power spectra.
- Other studies seek to locate particular clusters, filaments or voids.



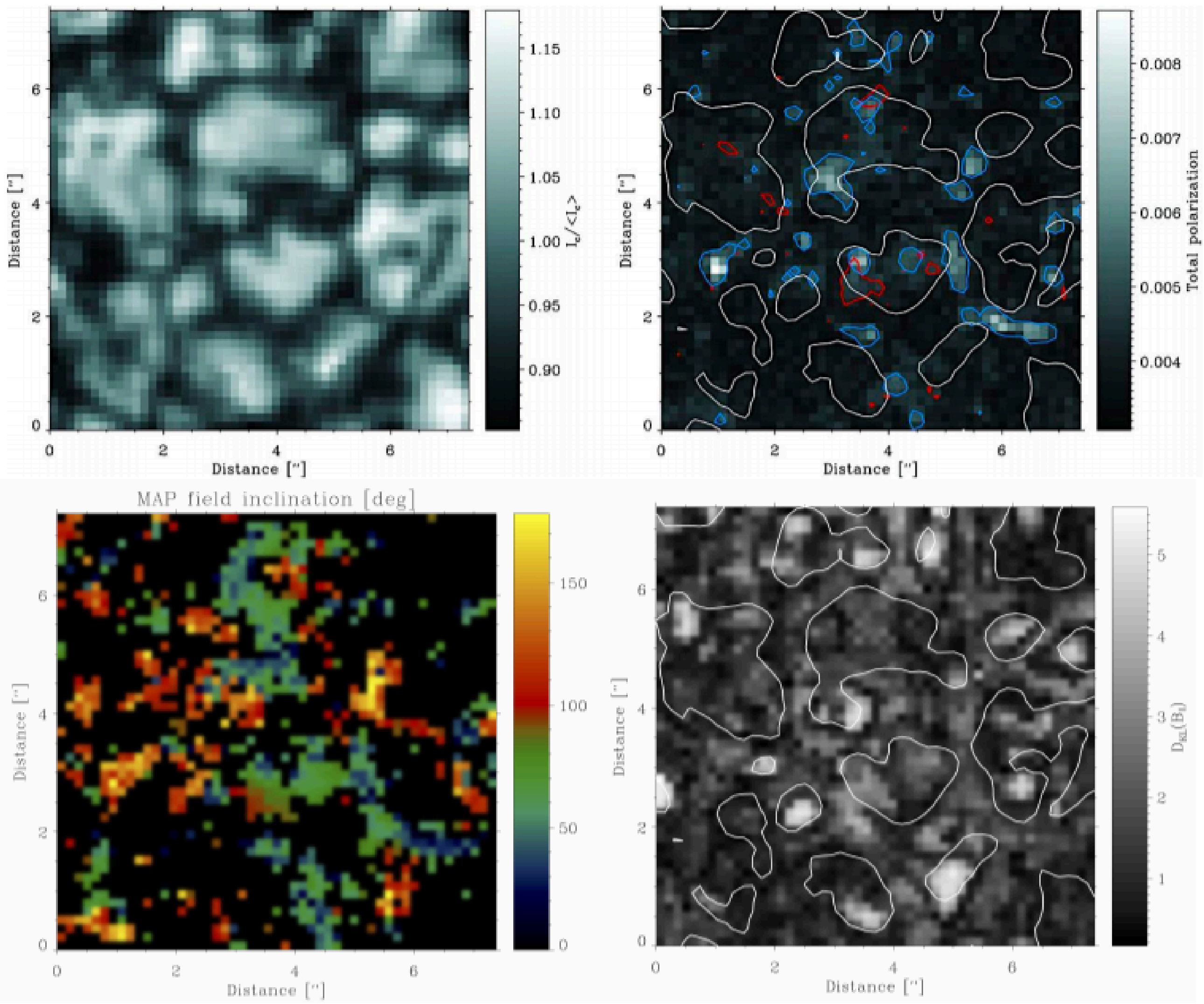
The photo-z conundrum

- Photometric redshift (photo-z) estimation has become a vital tool in the extragalactic astronomy and observational cosmology.
- The challenge of photo-z accuracy then depends on the statistical procedures used to calibrate photometric measurements to spectroscopic redshifts.



Bayesian modeling

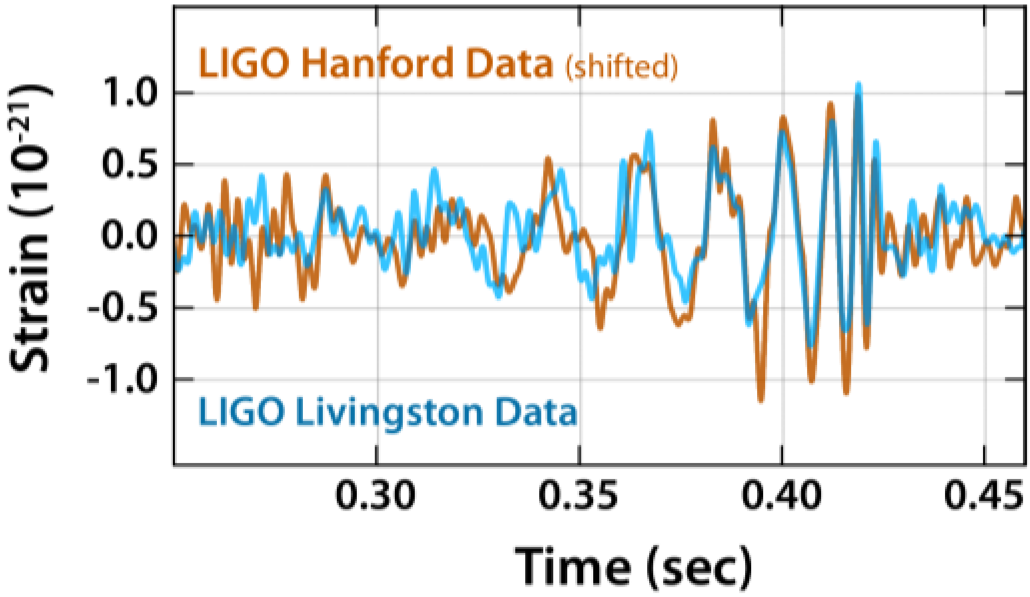
- Bayesian modeling has become a standard practice in many fields.
- Sampling algorithms, theoretical advancements related to prior selections, are all active research areas.
- Likelihood-free modeling: two main forms of statistical models can be distinguished: those describe by probability distributions for which an explicit likelihood can be written, and implicit or generative models.



Challenges in signal analysis

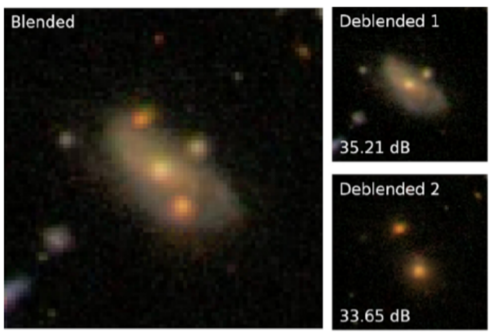
- The study of variable objects in the sky – time domain astronomy – is burgeoning with more than 2000 studies annually.

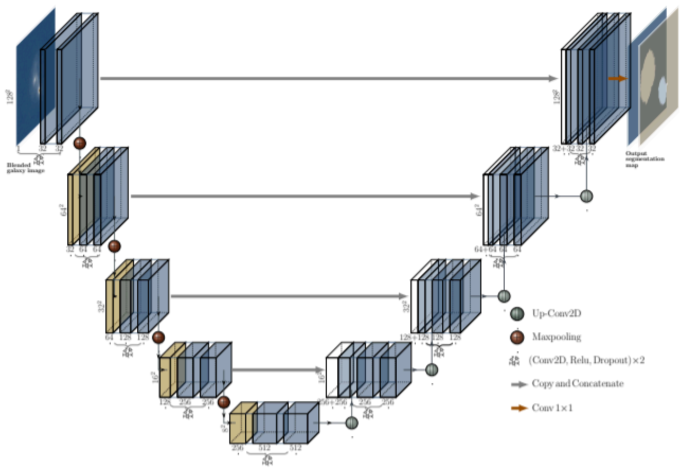
- Gravitational wave detection: the statistical challenge with is to detect short-lived chirp-like events in a continuous time series where noise is dominated by instrumental effects that can be continuous (perhaps caused by vibrations in the mirror structures) or transient (perhaps caused by minor Earth tremors).



Machine learning techniques

- This is literally an “exploding” field.
- Just a couple of instances: photometry of blended galaxies and the accelerated expansion of the universe.





AI tools

- Apart from the folklore around the subject, large-langage model technologies can have a strong impact on our researches, although still to largely be evaluated.

Final remarks

- Contemporary astronomical data analysis often elude the capabilities of classical statistical techniques, and inevitably requires the use and development of sophisticated, and sometimes novel, statistical tools.
- Astronomy requires expertise in vast fields of statistics and information science: nonparametric and parametric inference (especially Bayesian), high-dimensional nonlinear regression, censoring and truncation, measurement error theory, spatial point processes, image analysis, time series analysis, multivariate analysis, clustering and classification, and many other forms of machine learning.

Take all of this very seriously!

A forza di guardare il cielo e di respirare a pieni polmoni l’aria fresca della notte, mi sembrava di riempirmi di stelle

Tiziano Terzani

Reference & Material

Material and papers related to the topics discussed in this lecture.

- Feigelson et al. (2021) - "21st Century Statistical and Computational Challenges in Astrophysics"

Further Material

Papers for examining more closely some of the discussed topics.

- Nguyen et al. (2023) - "AstroLLaMA: Towards Specialized Foundation Models in Astronomy"
- Webb & Goode (2023) - "An Astronomers Guide to Machine Learning"

Course Flow

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html	Science case about CO_2 content in atmosphere	-	Course Summary

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